

Weekly internship Reflection

University of Niagara Falls

Master's in data Analytics

Internship information

Intern name: Abigia Gashahun Edossa

Student Number: NF1021072

Company Information

Company name: Food Truck Association of Canada (FTAC)

Internship job title: Data Science & AI pipeline engineer Intern

Internship Supervisor: Mackenzie Peck

Week 4 reflection: 2nd February -6th February, 2026

During Week 4, I mainly focused on preparing the dataset and building the basic data processing steps that are needed before creating a machine learning model. In simple terms, I worked on organizing the data so that a computer can understand and learn from it. Earlier in the project, I was having difficulty because the system was classifying too many posts as mixed help instead of clearly separating them into startup help or operational help. I also noticed that operational help posts were not being identified properly, even though I knew there were many of them in the dataset. Because of this problem, I decided to improve the way the data was prepared and processed, which is why I followed these steps.

One of the first things I did was set up automatic keyword extraction using TF-IDF. This method helps the system find important words and phrases from each Reddit post based on how often they appear and how meaningful they are across all posts. Instead of manually creating keyword lists, this approach lets the data speak for itself, which makes the system more accurate and scalable. This was an important step because it helped the model start recognizing patterns between different types of posts, such as posts from people trying to start a food truck versus posts from people who already run one and need help solving problems.

I also worked on organizing the categorical information in the dataset. I extracted all the unique subcategories that exist in the data and assigned each subcategory a unique numeric ID. The numbers themselves don't have meaning — they simply help the computer process the categories more efficiently. After assigning IDs, I created one-hot encoded columns for these subcategories. This means I converted each category into a yes/no format (1 or 0), which is the format machine learning models usually work best with. This step ensures the model can clearly understand which subcategories are present in each post.

In addition to this, I spent time cleaning and standardizing the dataset. I combined all relevant text columns into one column called `combined_text`, which makes it easier to process the text in one place. I also created a new column called `ml_label`, which classifies each post as either

startup-related or operational. This helps prepare the dataset for training a classifier later. By the end of the week, I had successfully cleaned, structured, and encoded the dataset. At that point, the data was ready to be used for machine learning model development, which will be the focus of the next stage of the project.

Student signature:

A handwritten signature in black ink, consisting of stylized, overlapping loops and strokes, likely representing the initials 'AS'.

Date: 07-02-2026